

SET THEORY (Chapter 1)

LESSON 1 - Set Notation

VOCABULARY

Set: A collection of distinguishable objects

ex. The set of whole numbers is $W = \{0, 1, 2, 3, \dots\}$

Element: An object in a set

ex. “3” is an element of the whole numbers

Universal Set: A set of all elements under consideration for a particular context; also called the sample space

ex. The set of integers is $I = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

Subset: A set of elements all belong to another set

ex. $O = \{1, 3, 5, 7, 9\}$ is a subset of D , digits 1 to 10

Notation: $O \subset W$ (on formula sheet)

Complement: All the elements of a universal set that do not belong to a subset of it

ex. $O' = \{2, 4, 6, 8, 10\}$ is the complement of $O = \{1, 3, 5, 7, 9\}$, a subset of the universal set of the digits from 1 to 10, D .

Notation: O' “prime sign” (on formula sheet)

Empty Set: A set with no elements

ex. The set of odd numbers divisible by 2.

Notation: $\emptyset$ or $\{\}$ (on formula sheet)

Disjoint: Two or more sets having no elements in common

ex. The set of even numbers and odd numbers are disjoint.

Consider the English alphabet; it is made up of consonants and vowels. We can organize this set by using the following:

All Letters: $U = \{$

Vowels: $V = \{$

Consonants: $C = \{$

In the above, set U is called the _____ set because it includes _____ possible elements.

- What can be said about the elements in set C in relation to those in set U ?
- What can be said about the elements in set V in relation to those in set U ?

When all elements in one set are included in another set, we say that one set is a _____ of the other set, and is denoted by the symbol _____. So, in our example we can write _____ and _____.

- What can be said about all of the elements in set C and set V ? Are there any elements in both these sets?

When none of the elements in one set are included in another set, we say that the sets are _____. If we wrote out the number of elements in this set, we have the _____ set, denoted by _____ or _____.

- Write all of the elements that are not in set C . _____

The elements of this set are called the _____ of set C and are denoted by _____.

- What would happen if we added the elements in set C and the set C' ?

So, whenever we add a set and its complement, we will receive the _____ set.

- How many elements are there in set V ? _____. When asked to write the number of elements in a set, the textbook may use the notation _____.

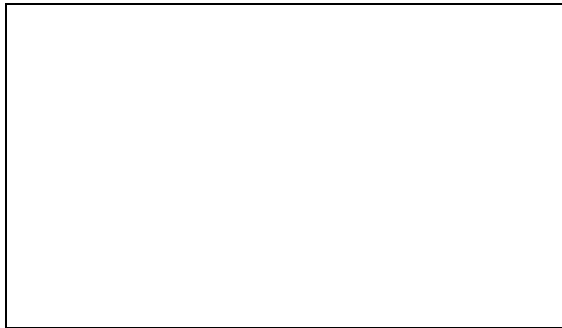
An extremely easy way to organize sets is to use a Venn diagram. Illustrate sets C and V below.



EXAMPLE 1

Consider the set of whole numbers, W , from 0 to 10. Let E be the set of even numbers, let O be the set of odd numbers, and let N be the natural numbers.

- $E = \{$
- $O = \{$
- $N = \{$
- Represent the information in a Venn Diagram



- List the complement of set E .
- List any disjoint sets.
- True or false?

$$E \subset N \text{ ______}$$

$$E' \subset W \text{ ______}$$

$$O' \subset E \text{ ______}$$

$$E \subset W \text{ ______}$$

$$W \subset N \text{ ______}$$

EXAMPLE 2

Consider the set of Prime numbers and Composite numbers. **Describe** these sets.

EXAMPLE 3

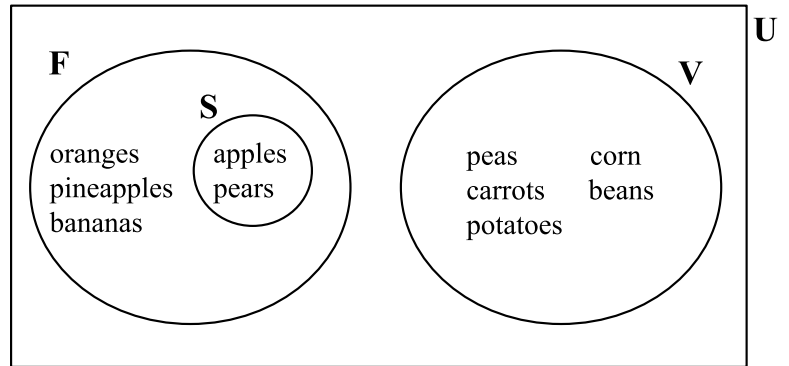
Johan drew the following Venn diagram and described the sets as follows:

$P = \{\text{produce}\}$

$F = \{\text{fruit}\}$

$V = \{\text{vegetables}\}$

$S = \{\text{fruit you can eat without peeling}\}$



a) The number of elements in V .

b) List any disjoint sets.

c) Is F' equal to V ? Explain.

d) List the elements in S'

Summary

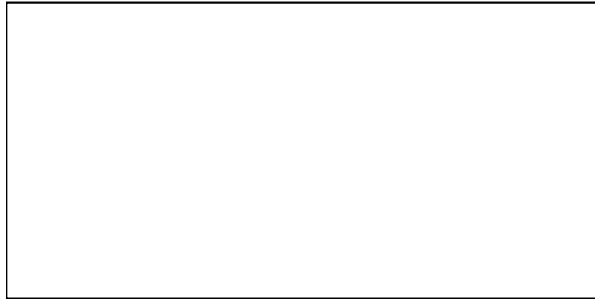
- Braces $\{ \}$ are used to define a _____.
- The symbol $\subset$ means one set is a _____ of another set.
- A _____ diagram is an easy way to graphically show sets.
- Two sets are called _____ when they have no elements in common.
- The complement, A' , is the set of elements _____ in the set A .
- All elements of all sets gives us the _____ set.
- The textbook uses the notation $n(A)$, which means we are to find the _____ elements in set A .

Practice

1. Using the following three sets

- $F = \{\text{first five multiples of 4}\} = \{4, 8, 12, 16, 20\}$
- $E = \{\text{first five multiples of 8}\} = \{8, 16, 24, 32, 40\}$
- $S = \{\text{first five multiples of 17}\} = \{17, 34, 51, 68, 85\}$

a) Draw a Venn diagram to represent these sets.



b) List the disjoint sets.

c) **Explain** whether or not each statement is true or false. If false, state why.

i. $E \subset F$ _____

ii. $F \subset E$ _____

iii. $E \subset E$ _____

2. Consider the following sets

$$W = \{\text{whole numbers}\}$$

$$N = \{\text{natural numbers}\}$$

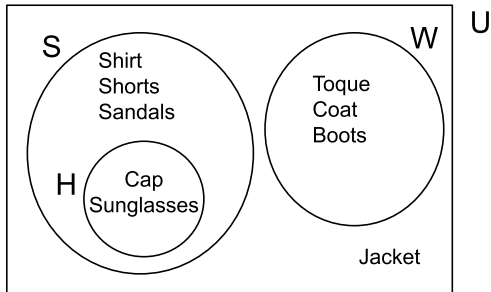
$$C = \{\text{composite numbers}\}$$

$$P = \{\text{prime numbers}\}$$

a) List all sets that are disjoint.

b) **Explain** if $W \subset N$ is true.

3. Consider the following Venn diagram.



a) A student made the statement that $S \subset H$. **Explain** if they are correct or incorrect.

b) State the number of elements in H' .

c) State the number of elements in W' .

d) **Explain** if S' equal to W .

e) List the disjoint sets.

Use the following information to answer the next question.

A student is working with a universal set that contains 52 elements that are all contained in two disjoint sets, A and B . The student determines that there are 13 elements in set A .

4. **Determine** the number of elements in set B .

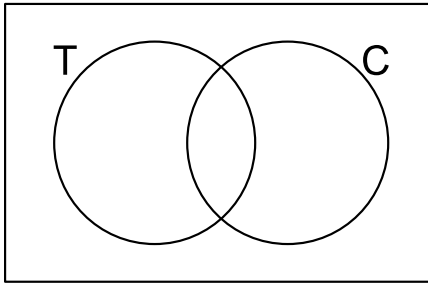
Use the following information to answer the next question.

A student is working with the set B and knows that B has 20 elements and B' has 31 elements.

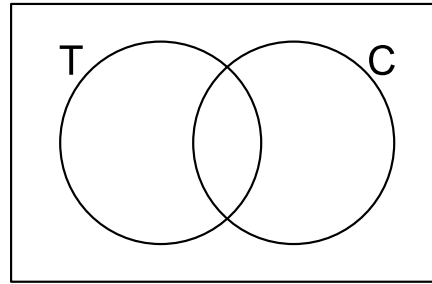
5. The number of elements within the universal set are _____.

6. Shade the appropriate region of the Venn diagram for the question given.

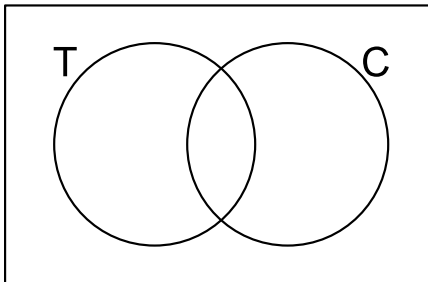
A. C'



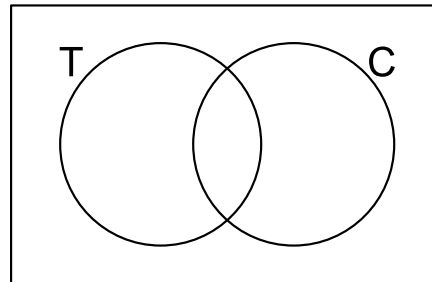
B. T'



C. Elements in T only



D. Elements in C only



7. a) Organize the following sets of numbers in a Venn diagram:

$$U = \{\text{integers from } -10 \text{ to } 10\}$$

$$A = \{\text{positive integers from } 1 \text{ to } 10\}$$

$$B = \{\text{negative integers from } -10 \text{ to } -1\}$$



b) List the disjoint sets, if any.

c) **Explain** whether or not each statement is true or false.

i. $A \subset B$

ii. $B \subset A$

iii. The number of elements in set A is equal to the number of elements in set B .

LESSON 2 - Venn Diagrams - Intersection and Union of Two Sets

Consider the following two sets and represent them in a Venn diagram

$A = \{\text{natural numbers less than 20 that are divisible by 3}\}$

$B = \{\text{natural numbers less than 20 that are divisible by 5}\}$



a) List the elements that are in both set A **and** also in set B .

A and $B =$ _____

The **INTERSECTION** of the two sets is the set A **and** B .

The word “and” in mathematics means the _____ of sets.

It is the set of elements which are members of both sets A and B ; that is, the elements that are in set A **and also** in set B .

It is the set $A \cap B$.

b) List the elements that are in set A or in set B .

A or $B =$ _____

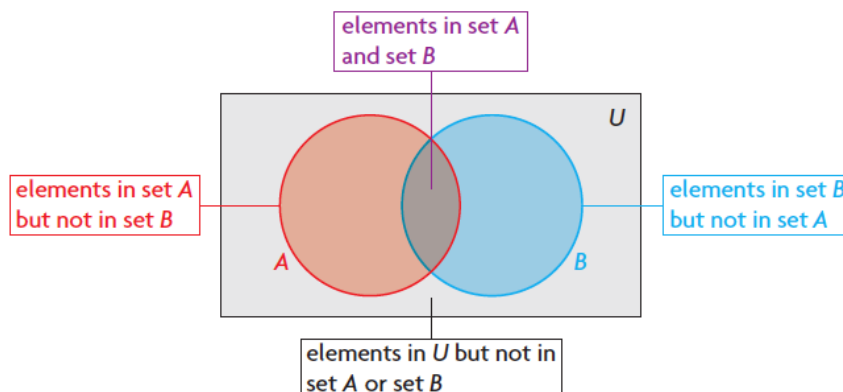
The **UNION** of two sets is the set of all elements in set A , **or** in set B , **or** in both.

The word “or” means the _____ of sets.

It is the set of elements which are members of both sets A or B ; that is, combining both sets.

It is the set $A \cup B$.

Venn Diagram

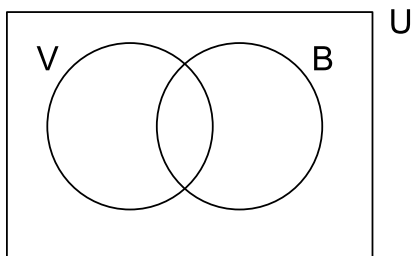


Organizing Data into a Venn Diagram

- 1) Determine the data that is both in Set A and Set B
- 2) Determine the data that is disjoint
- 3) Organize the numbers into each section of the diagram, check to see if the numbers match the total

EXAMPLE 1

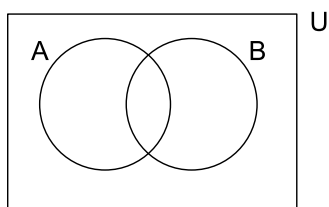
In an Alberta school, there are 65 grade 12 students. Of these students, 23 play volleyball and 26 play basketball. There are 31 students who do not play either sport. Using this information, complete the Venn diagram below.



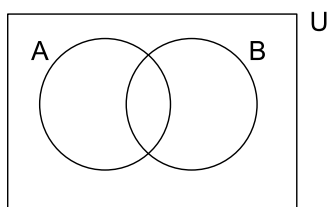
EXAMPLE 2

In each of the following Venn diagrams, shade the region representing the given set.

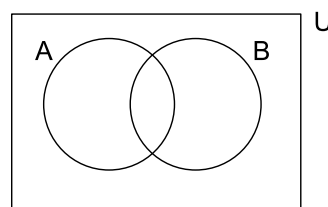
$A \cap B$ (A and B)



$A \cup B$ (A or B)



A' (Not A)



EXAMPLE 3

Use the Venn diagram to list the members of the following sets.

a) $P =$

b) $Q =$

c) $P \cap Q =$

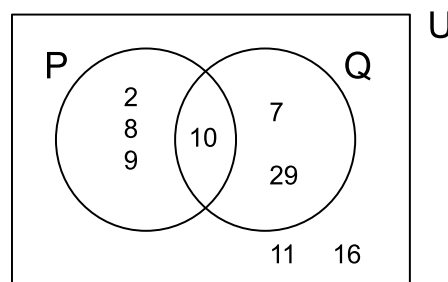
d) $P \cup Q =$

e) $P' =$

f) $Q' =$

g) $(P \cup Q)' =$

h) $(P \cap Q)' =$



Practice

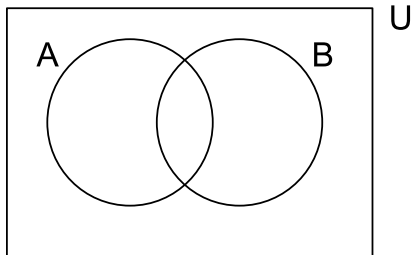
1. Consider the following sets:

The universal set $U = \{2, 3, 4, 6, 8, 9, 10, 12, 14, 15\}$

$A = \{3, 6, 9, 12, 15\}$

$B = \{2, 4, 6, 8, 10, 12, 14\}$

- a) Illustrate these sets using a Venn diagram.



- b) Indicate the number of elements in:

i. Set A: _____

ii. Set A but not in set B: _____

iii. Set B: _____

iv. Set B but not in set A: _____

v. Set A and Set B: _____

vi. Set A or Set B: _____

vii. Not in A: _____

- c) Express the following in set notation ($\cap$, $\cup$, X').

i. Set A: _____

iii. Set B: _____

v. Set A and Set B: _____

vii. Not in A: _____

ii. Set A but not in set B: _____

iv. Set B but not in set A: _____

vi. Set A or Set B: _____

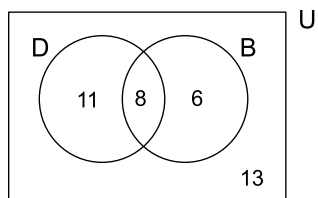
2. Consider the following sets:

$P = \{\text{the first five prime numbers}\}$

$E = \{\text{even numbers}\}$

Explain if $P \cap E = \emptyset$.

3. There are 38 students in a Grade 12 class. The number of students in the drama club (D) and the band club (B) are illustrated in the Venn diagram.



a) How many students are in both the drama club and in the band?

b) How many students are in the drama club but **not** in band?

c) How many students are in the drama club? How many are in the band?

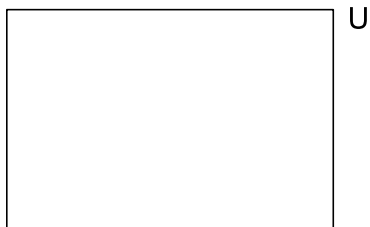
d) How many students are in at least one of the drama club or the band?

e) How many students are in not the drama club or the band?

4. Anna surveyed 45 students about the sports they play. She recorded her results below:

Sports Played	Number of Students
Hockey	20
Soccer	14
Neither hockey nor soccer	16

a) Draw and label a Venn diagram to show the data.

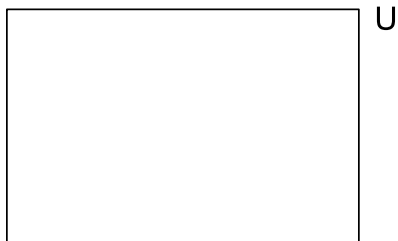


b) **Determine** how many students play hockey and soccer.

c) **Determine** how many students play only hockey or only soccer.

5. There are 55 guests at a ski resort. Of these guests, 25 plan to go skiing and 32 plan to go snowboarding. There are 9 guests who do **not** plan to ski or snowboard.

a) Draw a Venn diagram to illustrate this situation.



b) **Determine** how many guests plan to ski and snowboard.

c) **Determine** how many guests plan to only ski.

d) **Determine** how many guests plan to only snowboard.

Use the following information to answer the next question.

Tim was given three Venn diagrams with shading to represent a set operation. These diagrams are followed by four possible set operations.

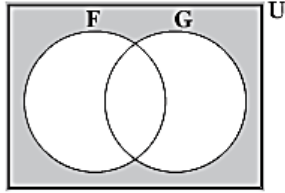


Diagram A

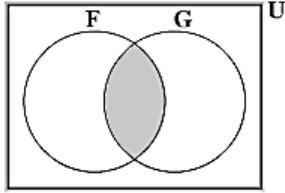


Diagram B

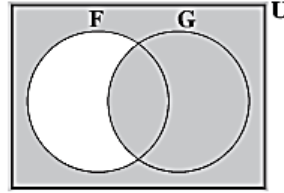


Diagram C

Set Operation	Reference Number
$F \cap G$	1
$F \cap G'$	2
$F' \cup G$	3
$(F \cup G)'$	4

6. Match each Venn diagram shown above with the reference number of the set operation that produces that diagram's shaded region. (use any reference number only once.)

Reference Number: _____

Diagram: _____

A

B

C

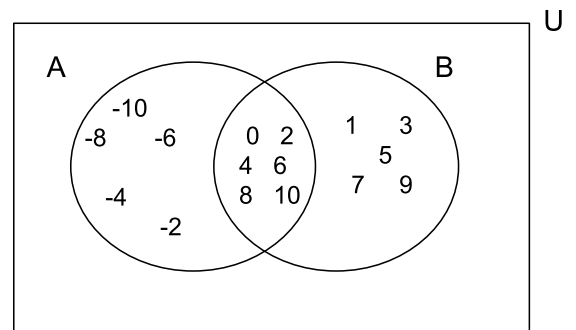
7. Consider the following Venn diagram:

a) Determine $A \cup B$.

b) How many elements are in $A \cup B$?

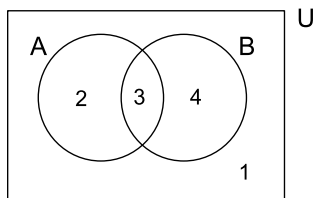
c) Determine $A \cap B$.

d) How many elements are in $A \cap B$?



Use the following information to answer the next question.

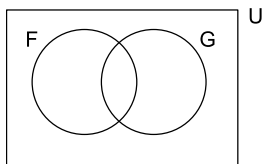
A Venn diagram is used to illustrate the relationship between sets A and B. The regions of the Venn diagram are numbered 1 through 4, as shown below.



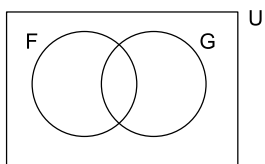
8. The regions of the Venn diagram that must be shaded to represent $(A \cap B)'$ are, in any order, _____

9. Given the Venn diagrams, shade the appropriate region(s) for

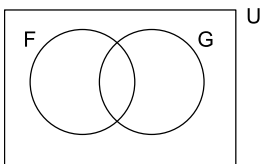
a) $F \cap G$



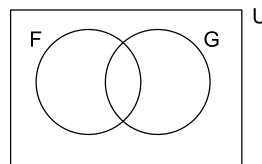
b) $F \cap G'$



c) $F' \cap G$



d) $(F \cap G)'$



10. A survey of 34 students showed that:

- 20 like Star Wars
- 15 like Avengers
- 4 like neither

Determine how many students like only the Avengers.

11. Darren asked 25 people at Comic-con if they liked G.I. Joe or Transformers

- 4 people did not like either movie
- 16 people liked G.I. Joe
- 11 people liked Transformers

Determine how many people liked both movies and how many liked only Transformers

12. Kevin asked 80 people if they liked the Oilers or Flames

- 9 people did not like either team
- 11 people liked both the Oilers and Flames
- 20 people liked only the Flames

Determine how many people liked only the Oilers.

13. Mario solved the following problem. A total of 48 students were asked how they got to school.

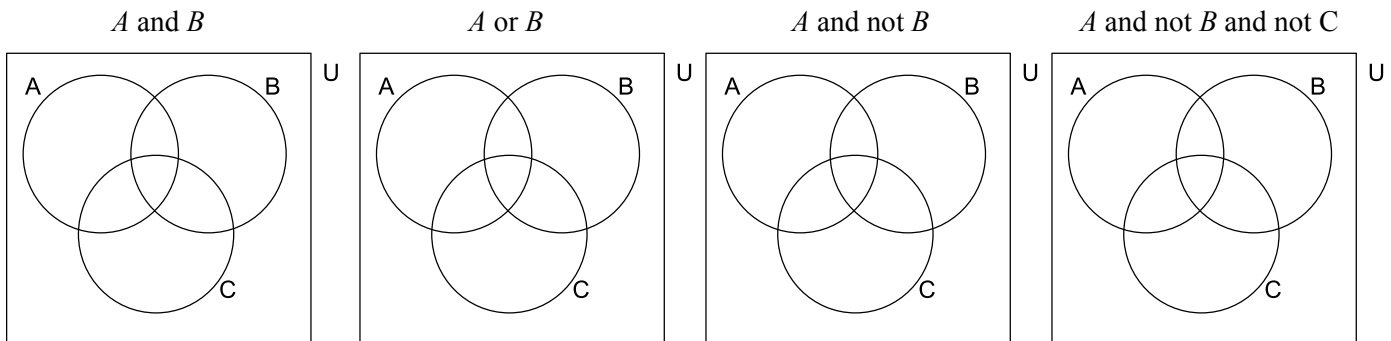
- 31 students drive a car
- 16 students take a bus
- 12 students do not drive a car or take a bus

Determine how many students do not take a bus to school.

LESSON 3 - Venn Diagrams - Intersection and Union of Three Sets

EXAMPLE 1

In each of the following Venn diagrams, shade the region indicated.

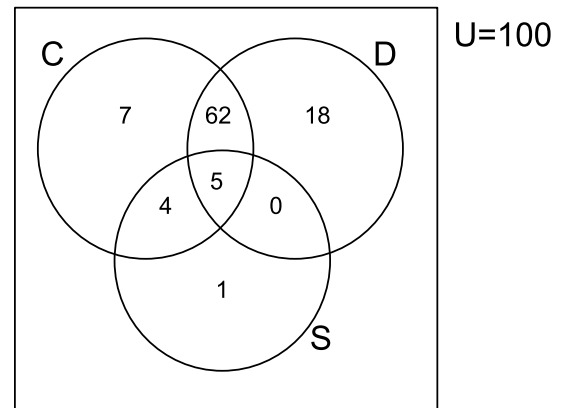


EXAMPLE 2

The Venn diagram on the right displays the results of a survey of 100 families regarding technology in their homes.

- C represents the number of families with a computer.
- D represents the number of families with a DVR.
- S represents the number of families with a scanner.

- a) What percent of the families have a computer at home?
- b) How many families have all three machines in their homes?

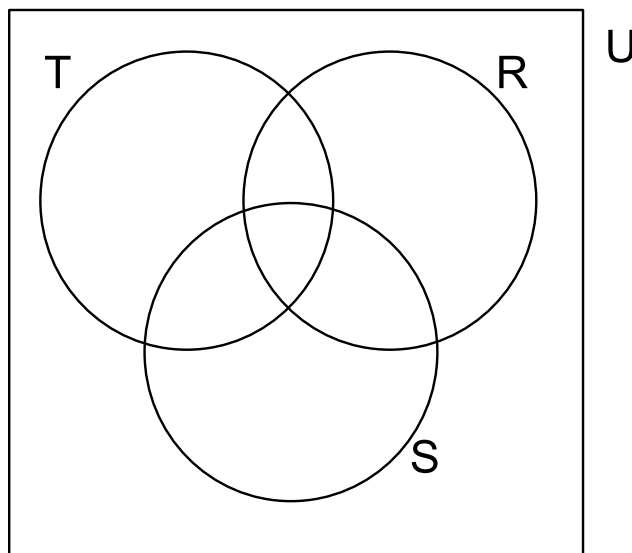


- c) How many families have none of the machines in their homes?
- d) How many families do not have a scanner machine?
- e) How many families have a computer and a DVR?

EXAMPLE 3

Each member of a sports club plays at least one of soccer, rugby, or tennis. The following information is known:

- 43 members play tennis,
- 11 play tennis and rugby
- 7 play tennis and soccer
- 6 play soccer and rugby
- 84 play rugby or tennis
- 68 play soccer or rugby
- 4 play all three sports.



a) Display the information in a Venn diagram.

b) How many members does the club have? _____

c) How many members play exactly one sport? _____

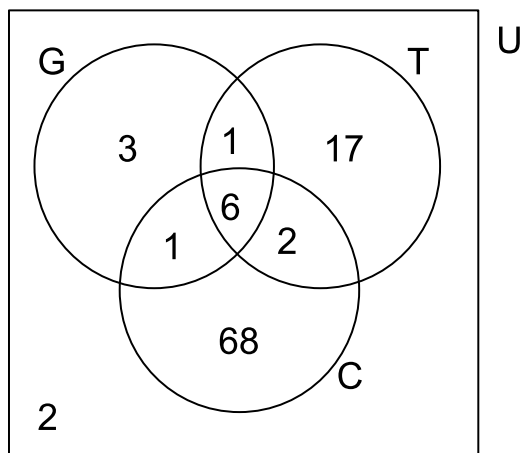
EXAMPLE 4

The Venn diagram represents the results of a survey of 100 students.

T = # students with a cell phone

C = # students with a computer

G = # students with a Gaming console



a) How many students own only a cell phone? _____

b) How many students own a cell phone? _____

c) How many students own a computer and gaming console? _____

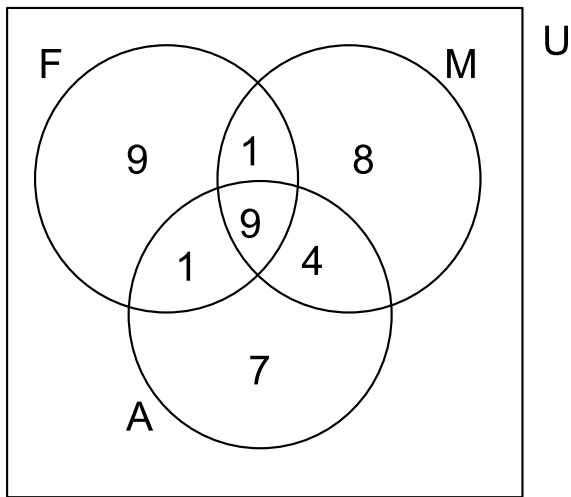
d) How many students own a computer or gaming Console? _____

e) What does the region $G \cap (T \cup C)'$ represent?

f) How many students own a computer or a gaming console but not a cell phone? _____

EXAMPLE 5

The members of a book club read fantasy (F), mystery (M), and adventure (A) books. The following Venn diagram shows the types of books that the members like.



Use the diagram to **determine** the number of people who read:

a) Fantasy or Mystery, but not Adventure _____

b) Adventure or Fantasy, but not Mystery _____

c) Neither Fantasy nor Mystery _____

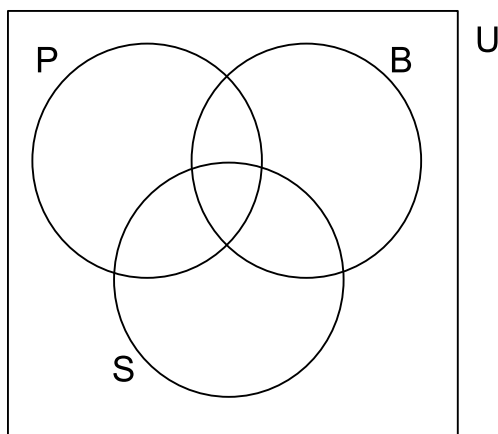
d) $(F \cap M)'$ _____

e) $(F \cup M) \cap A$ _____

f) $(A \cap F) \cup M$ _____

Practice

1. Did somebody say pizza? 80 people were surveyed about what meat topping they like on their pizza: Pepperoni (P), Salami (S), or Bacon (B). The results of the survey are shown below.



- 42 people said pepperoni
- 7 people said all three toppings
- 22 people said pepperoni and bacon
- 8 people said pepperoni and salami but not bacon
- 10 people said salami only
- 13 people said bacon and salami
- 4 people said they did not like any of the meat toppings.

a) **Determine** the number of people who like bacon only.

b) **Determine** the number of elements in each of the following:

$$P \cup B = \underline{\hspace{2cm}}$$

$$(P \cup S) \cap B = \underline{\hspace{2cm}}$$

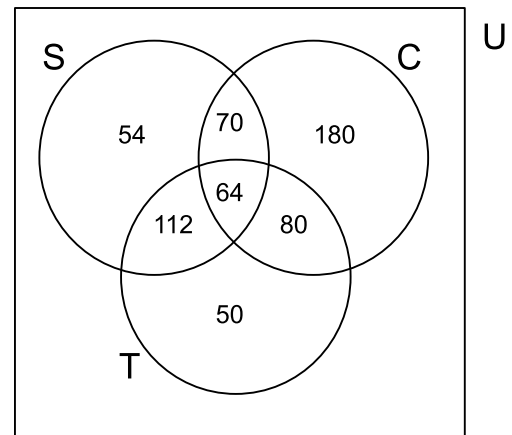
$$(P \cap B)' = \underline{\hspace{2cm}}$$

$$(P \cap B \cap S)' = \underline{\hspace{2cm}}$$

$$(P \cup B \cup S)' = \underline{\hspace{2cm}}$$

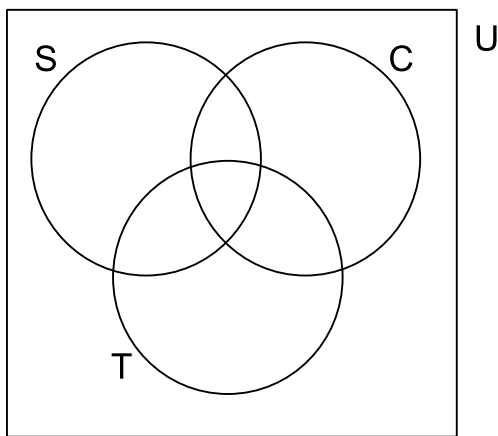
2. The city surveyed 300 people about where they get their morning coffee at Starbucks (S), Tim Hortons (T), and Second Cup (C).

- 54 went to Starbucks only
- 244 went to Tim Hortons or Second Cup
- 180 went to Tim Hortons
- 70 went to Starbucks and Second Cup
- 112 went to Starbucks and Tim Hortons
- 80 went to Second Cup and Tim Hortons
- 64 went to all three establishments



Babaganoush displayed the information in the Venn diagram above.

a) Help poor ol' Baba out by putting the correct numbers in the Venn diagram below.



b) How many people went to Second Cup and Tim Hortons but not Starbucks? _____

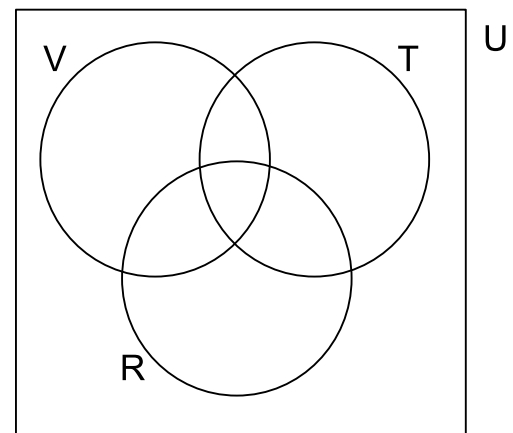
c) How many people went to Tim Hortons and Starbucks? _____

d) How many people went to Tim Hortons only? _____

e) How many people do not get morning coffee ? _____

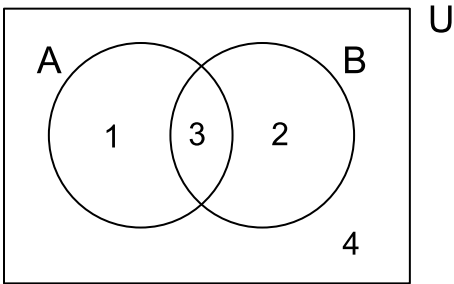
3. 100 people were surveyed about who owns the following super cool technologies: VCR (V), 8-track player (T), and Record (R) players.

- 5 owned none of the technologies
- 15 owned all three
- 5 owned a VCR and a Record player but not a 8-track
- 25 owned a VCR and a 8-track
- 12 owned a Record player only
- 45 owned a VCR
- 2 owned a 8-track and a Record player but not a VCR
- 83 owned a VCR or a 8-track



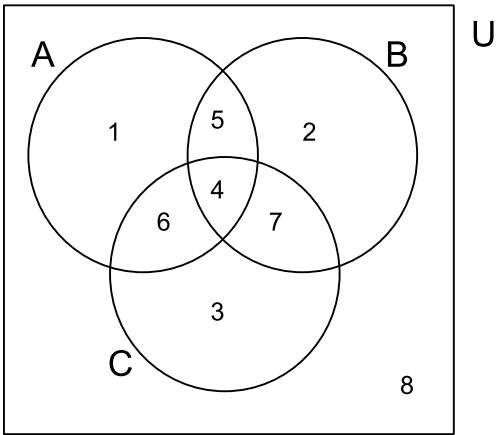
Determine how many people owned only an 8-track.

4. Babaganoush is given the following Venn diagram and is asked to determine the following regions. Find her mistakes and correct them.



Question	Babaganoushs' Answers	Correct Answer
Elements in A	region 1	
$A \cap B$	region 2	
Only B	region 3	
$A \cup B$	regions 1 and 3	
not A	regions 2 and 3	

5. Complete the table using the given Venn diagram. Identify the regions that represent the “Questions”.



Question	Region(s)
Only A	
$A \cap B$	
$A \cap B \cap C$	
$A \cup B$	
C'	
A and B and not C	
C or B not A	

Use the following information to answer the next **two** questions.

The following three sets are given:

$$R = \{\text{natural numbers less than } 50\}$$

$$S = \{\text{even numbers}\}$$

$$T = \{10, 20, 30, 40\}$$

6. Which of the following statements is true for sets R , S , and T ?

A. $R \subset S$

B. $R \subset T$

C. $S \subset R$

D. $T \subset R$

7. Which of the following statements is **not** true for sets R , S , and T ?

A. $T \subset (R \cap S)$

B. $T \subset (R \cap T)$

C. $(R \cap S) \subset T$

D. $(R \cap T) \subset T$

8. While researching new vehicles to purchase, a student found 80 cars that he was interested in. 31 cars had air conditioning only, 15 had leather seats, 41 had air conditioning. How many cars did not have neither air conditioning or leather?

9. A total of 40 teens were surveyed about their training preferences: Cycling or Swimming

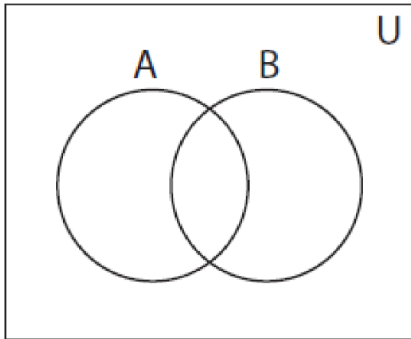
- 12 trained in both cycling and swimming
- 11 did not train in cycling or swimming
- 3 more trained in cycling only than swimming only

Determine the number of teens who trained in cycling only.

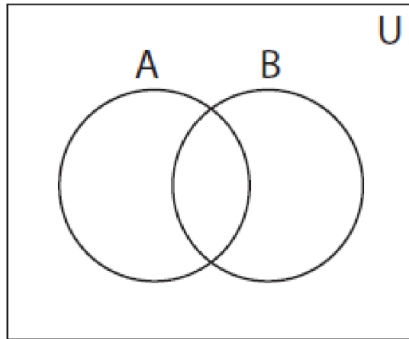
Set Theory Shading Practice

1. Given the Venn diagrams, shade the appropriate region(s) for

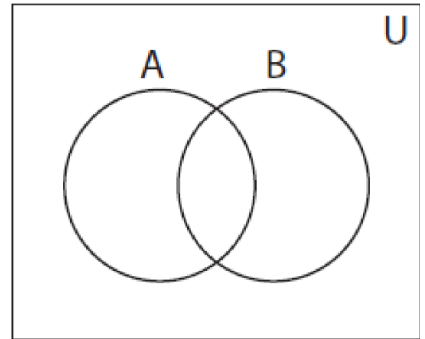
a) $A \cup B$



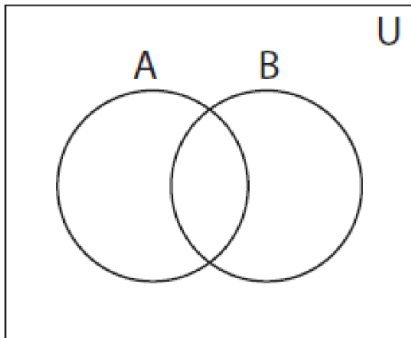
b) A



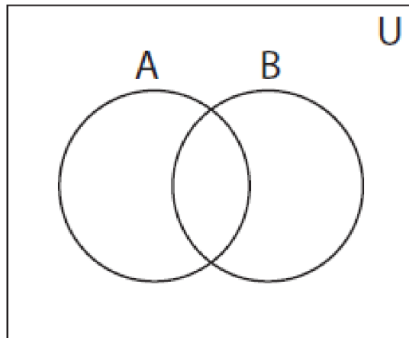
c) $A \cap B$



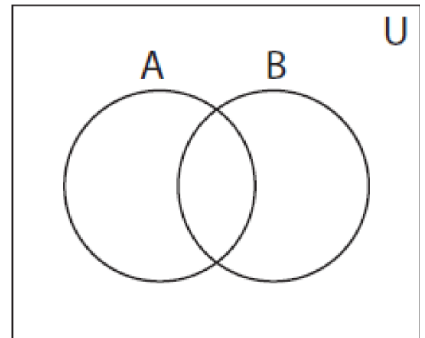
d) B'



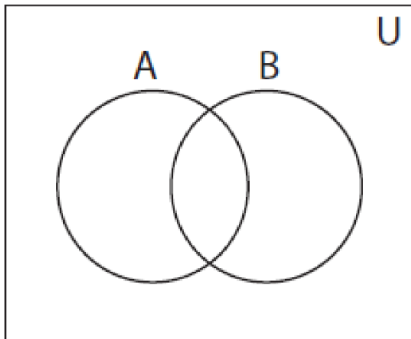
e) $(B \cup A)'$



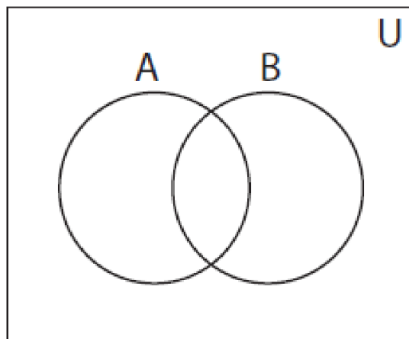
f) $A' \cap B$



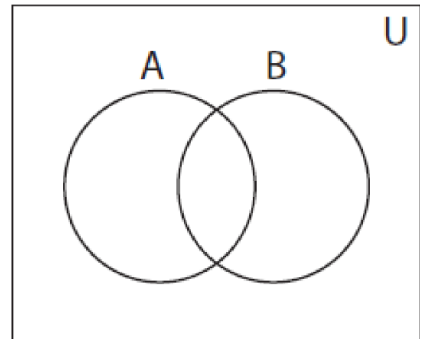
g) $A' \cup B'$



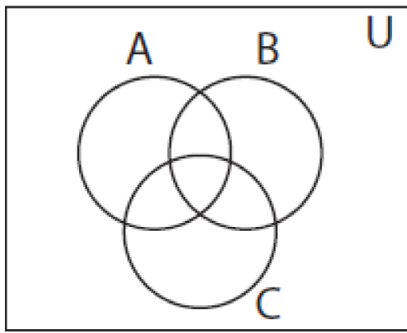
h) $A \cup B'$



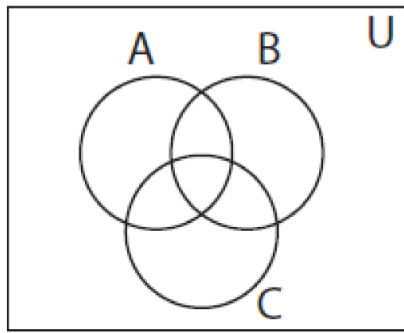
i) $(A' \cap B)'$



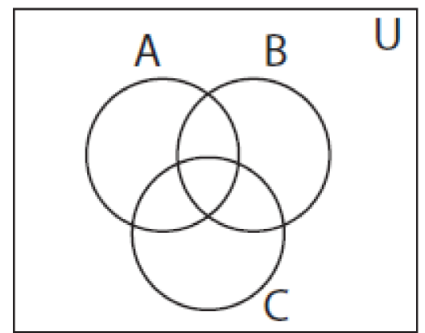
j) $A \cap C$



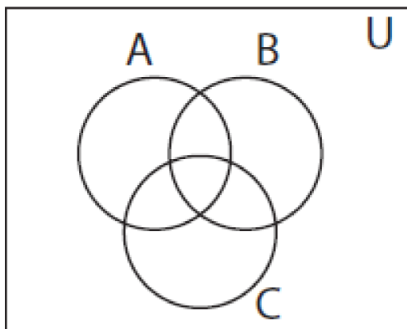
k) A'



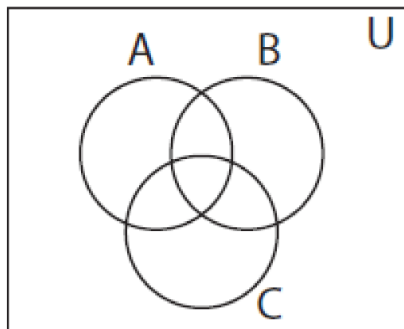
l) $A' \cup C$



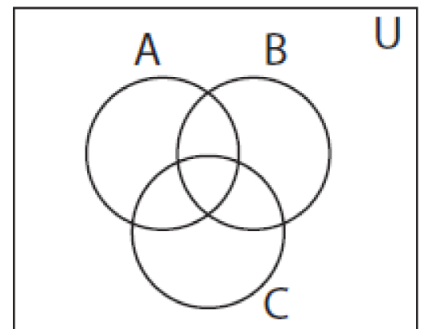
m) $A' \cup B'$



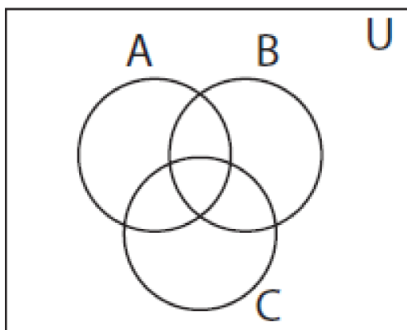
n) $(A \cap C) \cup B$



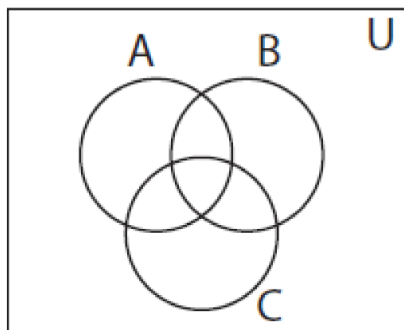
o) $(B \cup C) \cap A$



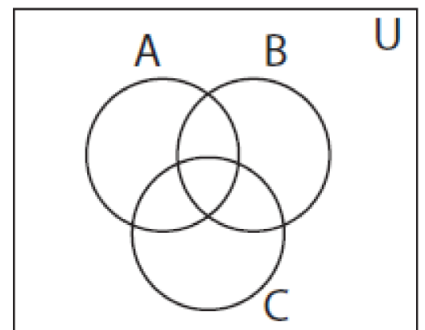
p) $A \cup B \cup C$



q) $(A \cup B) \cap C'$

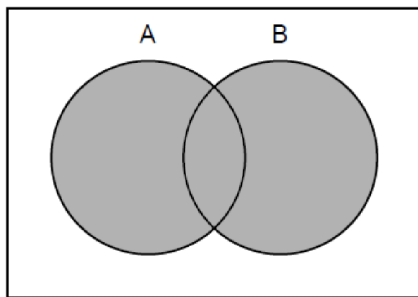


r) $(A \cap B) \cap C$

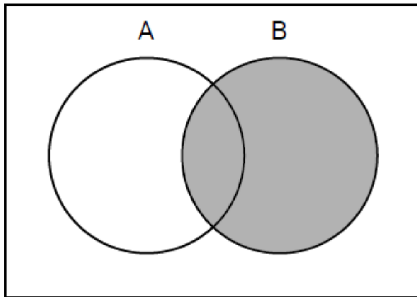


2. Name the shade region for the following Venn diagrams.

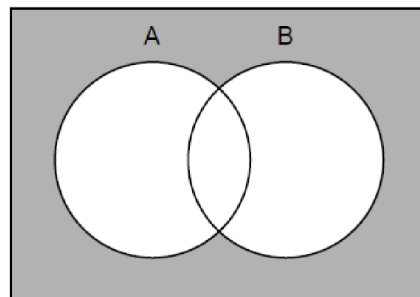
a) _____



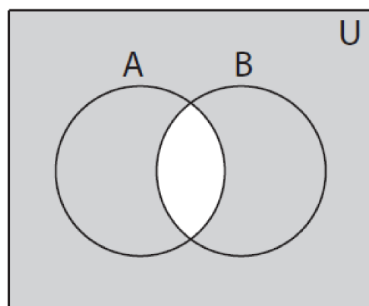
b) _____



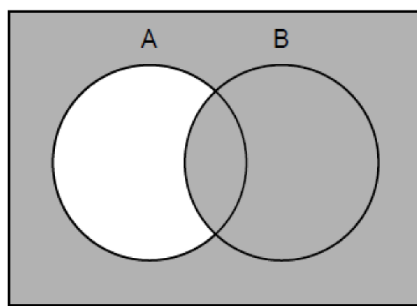
c) _____



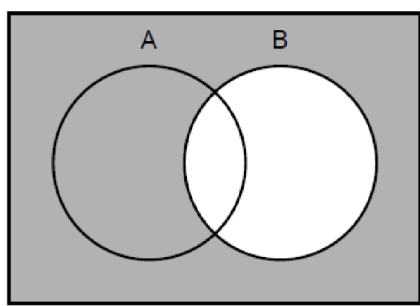
d) _____



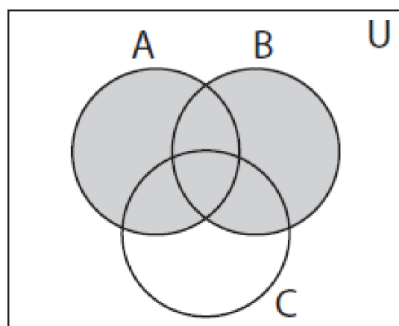
e) _____



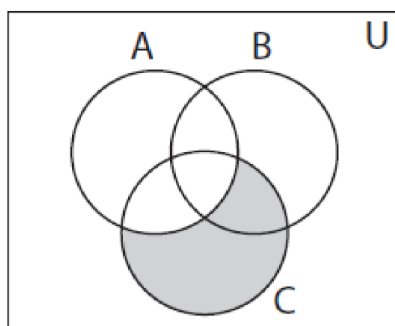
f) _____



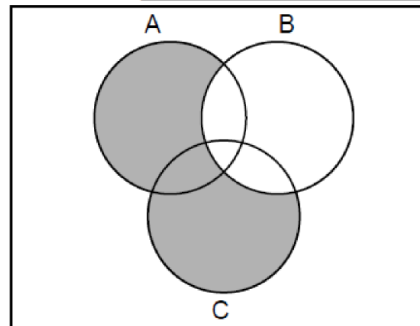
g) _____



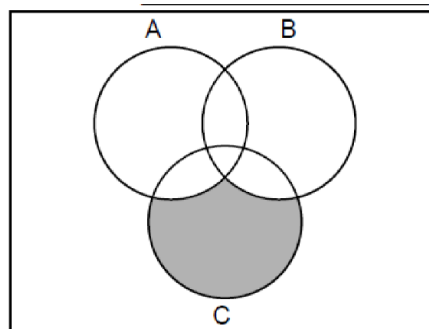
h) _____



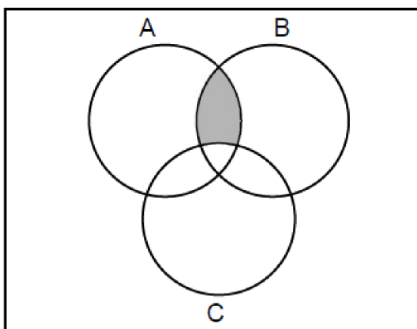
i) _____



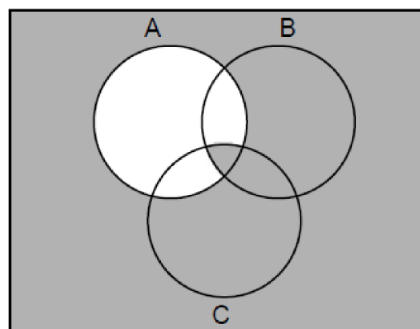
j) _____



k) _____



l) _____

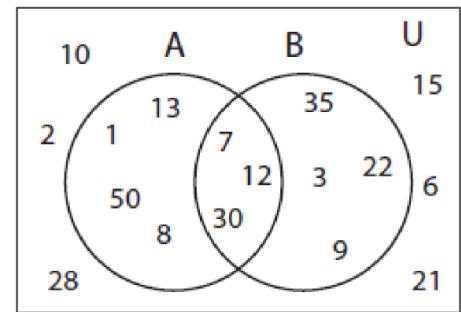


3. Consider the following Venn diagram:

$A \cup B$: _____

B' : _____

$A' \cap B$: _____

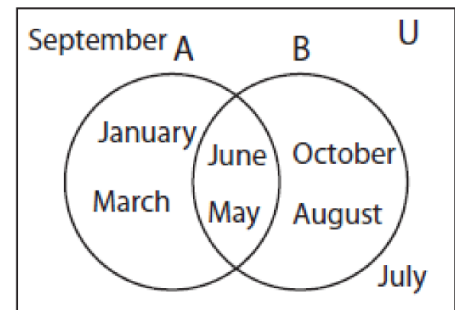


4. Consider the following Venn diagram:

$A \cap B$: _____

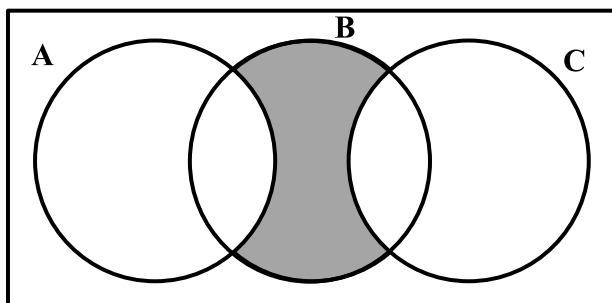
$A' \cup B'$: _____

$A \cup B'$: _____

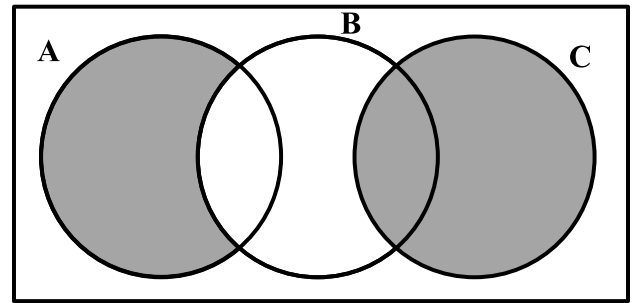


5. Name the shade region for the following Venn diagrams.

a) _____



b) _____



6. Given the Venn diagrams, shade the appropriate region(s) for

a) $A \cup (B \cap C)$

b) $(A \cup C) \cap B'$

